



Customer 360

Customer States

June 2026

Agenda

- 1. Progress**
- 2. States Framework** – alignment on the framework for the States.
- 3. Approach for defining behaviors**
- 4. Variable importance**
 1. RFMC features
 2. Behavior features
- 5. State Methodology**
- 6. Predictive Models Introduction**
- 7. What is next**

The purpose of today's meeting is to get alignment on the outputs and approach

We'll show you some data as well

This is an internal slide show for the analytics team and not designed for a wider audience. Much of the material will be difficult to understand without proper context and discussion.



High Priority



Delivery



Completed



On Track



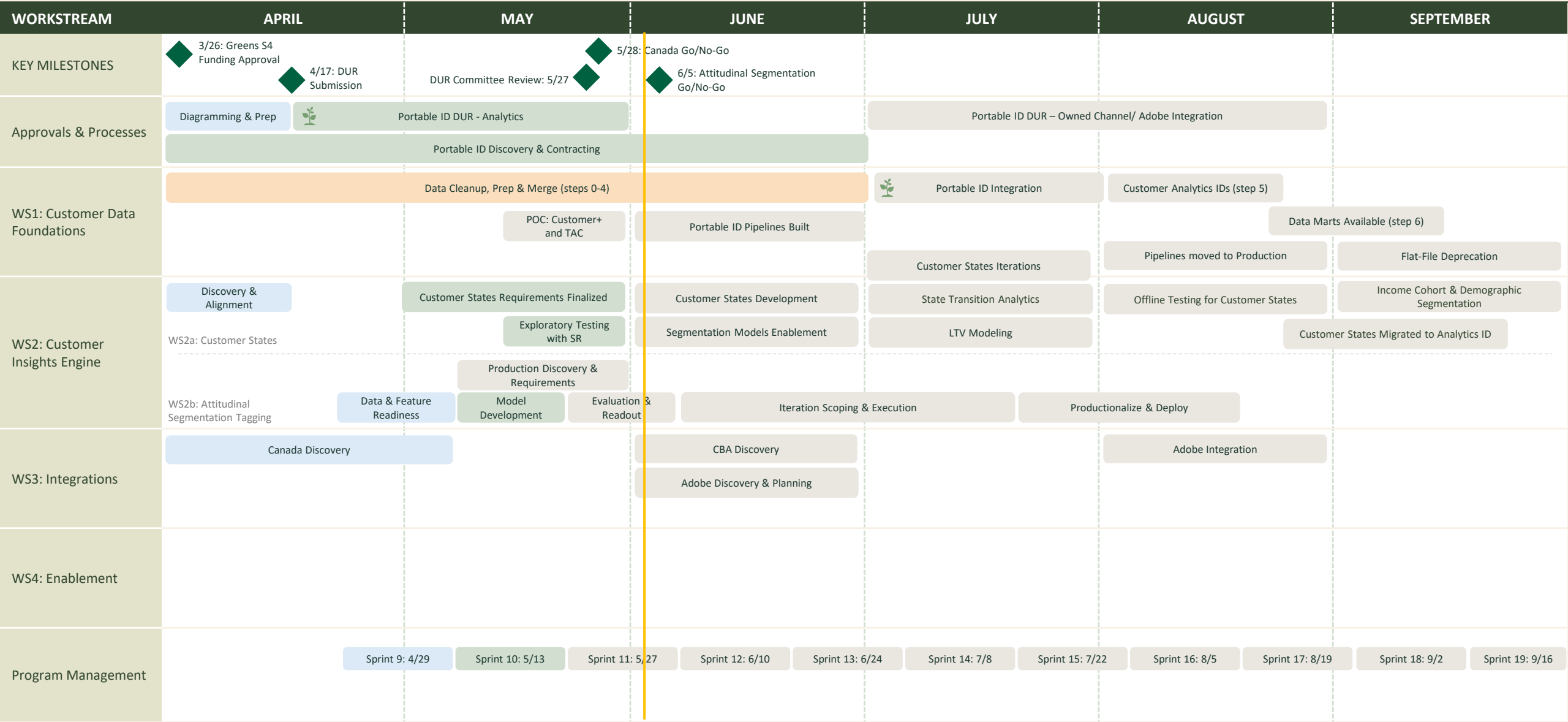
At Risk



Blocked



Not Started



Worried About

- I am worried about our **timelines**.
 - This type of work is highly *iterative*. So, we must move fast enough to provide you (and stakeholders) with results that you can react to and we can iterate on.
 - One month to create the States definition and one month to create the predictive models is highly aggressive.
- I am worried about **adoption**.
 - Not too worried because Sarah is on it. However, the initial meetings and comms to the stakeholder teams should be persuasive.
 - I would love to run some experiments prior to doing a lot of production work.
 - Reports are not on our roadmap
- I am worried about what this looks like in **production** and how we get there.
 - Data Science can get us to the point where we have “a table in Databricks that updates daily”
 - After that, *MLOPs, production-level data pipelines, and Adobe integration* are not on our roadmap.

States Framework

Segment ID*	Consistency	Recency	Shopping	Coupon	Channel	Shopper Description	Population %*	Engagement Desc.	Segment
1	highest	highest	highest	highest	highest	Top Engagers	5.3%	High	Most Val
2	highest	high	highest	high	highest	Loyal customers with high category engagement	2.1%	Medium	Stabl
3	high	high	high	high	highest	Stable tenure and recency with above avg. category and high interaction	10.6%		
4	high	high	high	Medium	Medium	Stable tenure and recency with above avg. category and lower interaction	11.2%		
5	Low	high	Medium	Low	Medium	Low tenure but high recency and avg. store depth- low coupon engagement	8.8%	Low	New Reactiv
6	Low	high	Medium	high	Medium	Low tenure but high recency and avg. store depth- above avg. coupon engagement	2.1%		
8	Low	high	Low	high	Medium	Low tenure but high recency and above avg. coupon engagement- low store depth	1.7%		
9	Medium	Medium	Medium	Medium	Medium	Middle of the road customers who are not new or stable but show promise in other interactions	1.3%		Declin
10	high	Medium	Medium	Medium	high	High tenure shopper but less recent and high communication interaction	1.8%		
11	high	Medium	Medium	Medium	Low	High tenure shopper but less recent and low communication interaction	3.7%		
20	highest	high	Low	Medium	Medium	High tenure shopper with high recency, but lack store depth	3.5%		
21	Medium	high	Low	Medium	Medium	Medium tenure shopper with high recency, but lack store depth	4.4%	Unengaged	Lipov
13	Low	Low	Medium	Medium	Medium	Low tenure and recency and engage with coupons	4.8%		
14	Low	Low	Medium	Low	Medium	Low tenure and recency but do not engage with coupons	8.8%		
15	lowest	Low	Low	Medium	Medium	Low tenure and recency with low engagement and little store depth	4.4%		
16	lowest	Low	Low	Low	Medium	Low tenure and recency with low/no engagement and little store depth	2.4%		
17	Low	Medium	Medium	Low	Low	Low tenure with mid recency and low channel/coupon engagement	12.9%		
18	lowest	lowest	Low	Low	Low	Low tenure and lack recency- some store and channel engagement	7.5%		Unenga
19	lowest	lowest	Low	lowest	lowest	Low tenure and lack recency- little to no store and coupon engagement	2.8%		
24	NA	NA	NA	NA	NA	Last activity was 12+ months ago, no longer would be tracked in data		NA	Churn

Behavior Dimensions Should Make Sense

Can be **conceptual**, but must be interpretable

Easy to Understand

Stakeholders should understand the behavior and what specific types of customer actions drive higher values of the behavior.

Emotionally Intuitive

Stakeholders should believe that the behavior is tied to value and that changes in customer actions will drive more value.

Operationally Actionable

Stakeholders should be able to identify levers that allow them to influence the behavior and thus drive more value

Strategic

The behaviors should be strategically tied to business initiatives.

What are the 3 behaviors?

Not all the behaviors meet our requirements

BEHAVIOR	EXAMPLES	ACTIONS
STAR and DISCOUNT BEHAVIOR <i>How do customers take advantage of SR, games, and other discounts?</i>	- Star usage - Recency & frequency of Star usage - Recency & frequency of discounts - Game participation	- Encourage STAR use - Design Star benefits Store Ops Marketing Loyalty
PRODUCT PREFERENCES <i>What products do customers prefer?</i>	- Product / category depth - Exclusive products - Cold/hot decisions	- Drive new product adoption - Product / category level recommendations - Advertising comms Marketing Product
VISIT STYLE <i>How customers interact with us day-to-day.</i>	- RFMC of channel use - Channel depth - Day part - Weekend	- Encourage diversity of channel use - Retain customers who no longer uses a channel - Encourage Peak visits Marketing Product Store Ops

I want to decide today, so that we are unblocked. However, we are creating a “methodology” that can be repeated for any three behavior types. So, we are not locked in.

How Many States

Twenty or so detailed States that roll up to 5-6 interpretable States

		VALUE SIGNALS		BEHAVIOR SIGNALS			
	State	Monetary	Recency	SR and Discount	Product Depth	Channel	Shopper Description
State							
1	Most Valuable	Highest	Highest	Highest	Highest	Highest	Highest spend frequency and deepest engagement across channels
2							
3							
4							
5	Stable	High	High	High	High	High	Consistent purchasers with strong product category engagement
6							
7							
8							
9	New & Reactivated	Low	High	Medium	Medium	Medium	Recently started or returned, with low purchase consistency so far
10							
11							
12							
13	Declining	Medium	Medium	Low	Low	Low	Previously consistent customers whose visit frequency is dropping
14							
15							
16							
17	Lapsing	Low	Low	Low	Low	Low	Low recent activity, at risk of disengaging entirely
18							
19							
20							
	Unengaged	N/A	N/A	N/A	N/A	N/A	No meaningful engagement in the current period

Approach for defining behaviors

Behaviors And Features

Behavior

Example: Channel
Use

Features

Customer level measures of Channel use that are intuitive:

- Number of channels used
- Diversity of channel use
- Recency of MOP transactions
- Frequency of MOP transactions
- AOV of MOP transactions
- ...

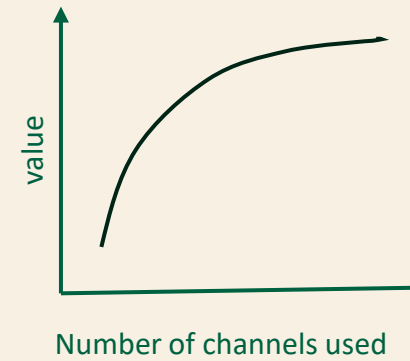
Select the features that matter

Feature selection to find the subset of features that:

1. Are correlated with future value
2. Are intuitive to our stakeholders
3. Are orthogonal to each other

Define the relationship

Define the relationship between the reduced number of features, which measure “Channel Use”, and future value.



Features – These are the metrics that are going to describe the behaviors:

- 1. RFMC**
- 2. Star and Discount**
- 3. Product Preferences**
- 4. Channel Use**

RFMC Features

We want **two** transactional features (based on RFMC) that are intuitive and correlate highly with value

State	VALUE SIGNALS	
	Monetary	Recency
Most Valuable	Highest	Highest
Stable	High	High
New & Reactivated	Low	High
Declining	Medium	Medium
Lapsing	Low	Low
Unengaged	N/A	N/A

- Easy to understand
- Intuitive so that stakeholders believe it
- Correlate or predict future value
 - Retention
 - Value

Methodology: Feature selection approach.

- Find the two features that predict **90-day spend (revenue)**.
- Use exhaustive feature list
- Use feature selection approaches
- Use R-squared to explain how much of the 90-day spend is explained by the features
- Bias toward orthogonal features

RFMC Features

We want **two** RFMC features that are intuitive and correlate highly with value

Concept	Feature	30-days	90-days	180-days	1 year
Recency	Days since last order				X
Frequency	Number of transactions	X	X	X	X
	Number of days with a transaction	X	X	X	X
	Number of weeks with a transaction		X	X	X
Monetary	Average monthly net revenue	X	X	X	X
	Log average monthly net revenue	X	X	X	X
Momentum	\Delta of frequency				
Consistency	$1/(1+CV)$, CV= coefficient of variation of inter purchase dates		X	X	X
	Entropy = $\sum(x * \ln(x))$ where $x = \text{day_diff} / \sum(\text{day_diff})$ per customer. Higher entropy → more evenly spaced transactions.		X	X	X
	second_moment = Second moment of the gap distribution		X	X	X
	log_utility = Log utility of the gap distribution: $\sum(\ln(x))$		X	X	X



Insights into RFMC Features

Value Exploratory data analysis.

Goal: identify which **transactional** features are best to include in the States Definition.

- **Intuitive** – explainable
- **Predictive** – predict future transactions
- **Simple** – include less than 3

Study Design

- Use feature engineering and selection to find best subset of possible **transactional** features.
 - Keep the potential features to a minimum
- *Average monthly spend* (Net discounted revenue) for 3 months as target. Log transformed.
- Target period: March – May 2026
- Feature period: June, 2025 – Feb, 2026 (lookback window)

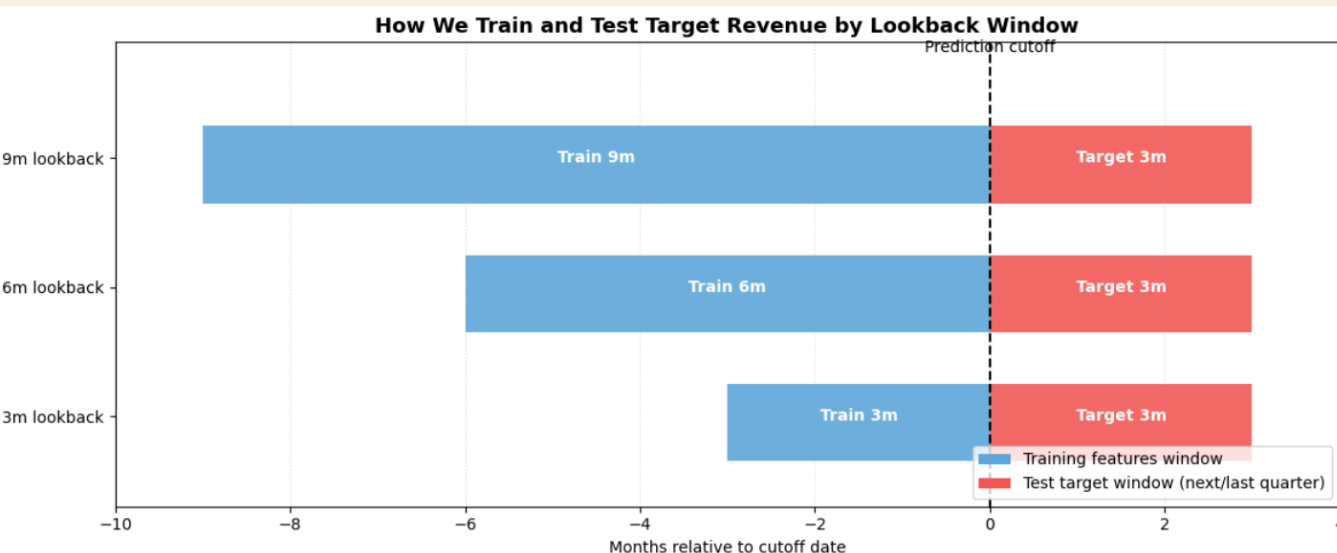
Random Sample of 500K customers to train.

Three models used to evaluate feature importance:

- Random Forest
- Ridge regression
- Forward / backward selection (linear regression)
- Gradient boosted machines with Shapely Values

Features

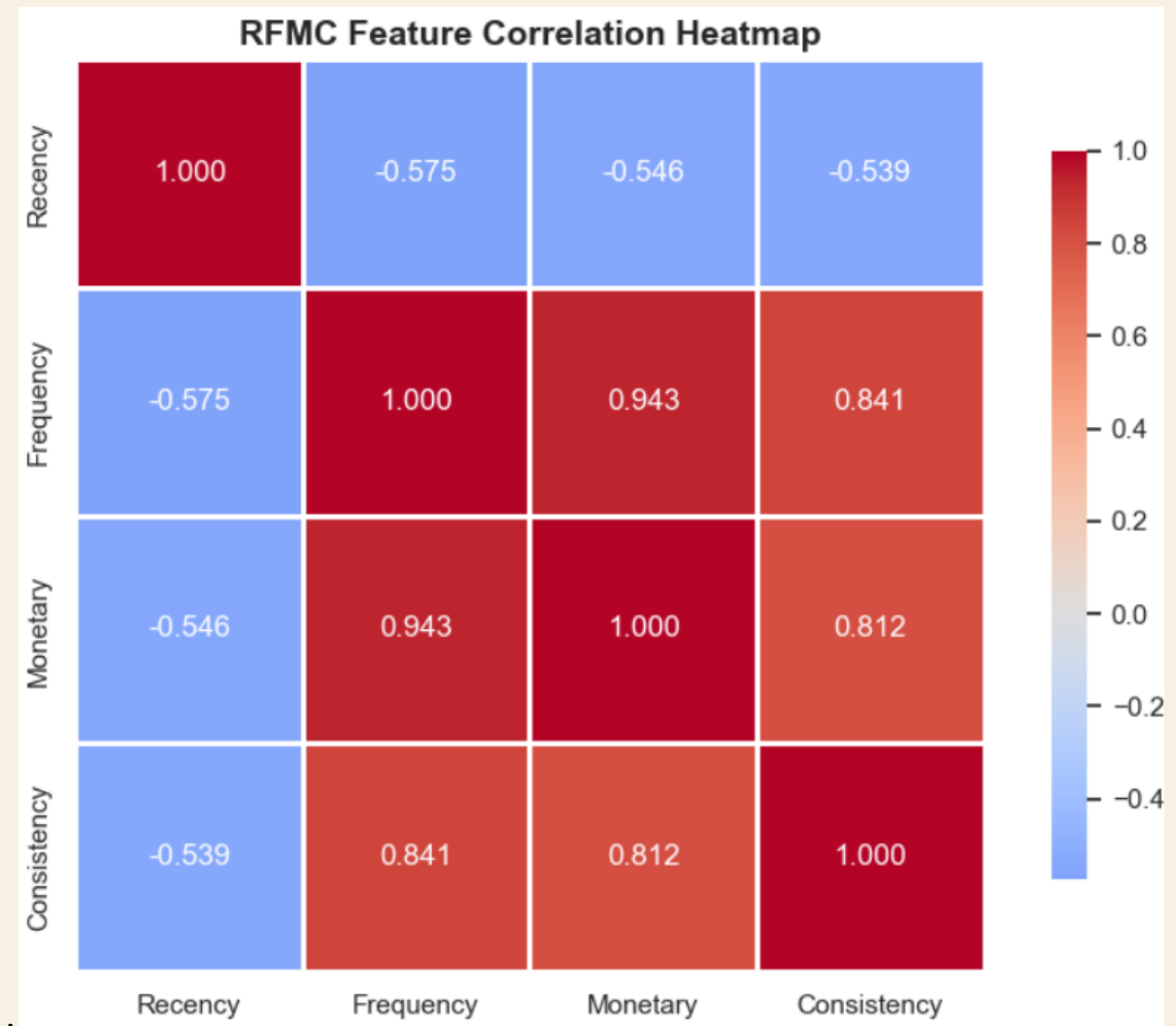
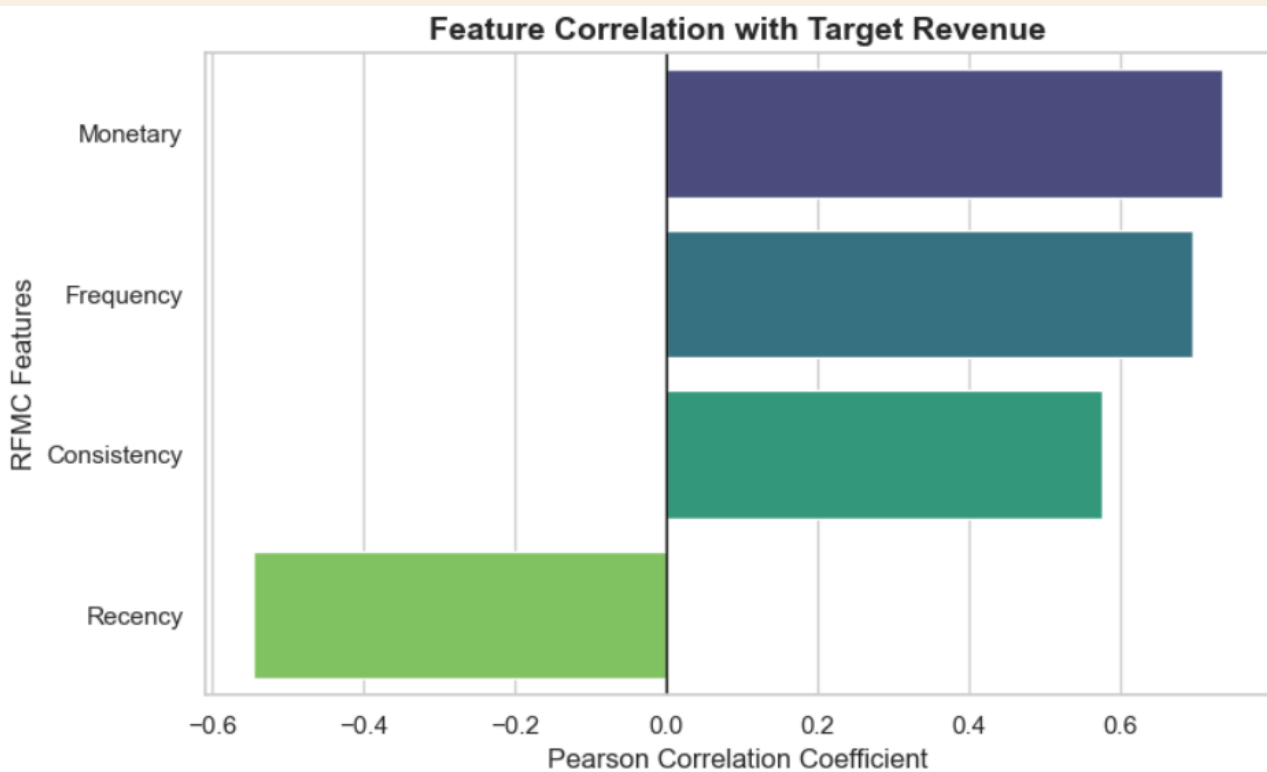
- **Recency (R)** : The time (days) since last purchase before the target period cutoff
- **Frequency (F)**: Total number of transactions made by the customer during the lookback window.
- **Monetary (M)**: Average monthly net revenue generated by the customer during the lookback window.
- **Consistency (C)**: [Entropy, log-utility, 2nd moment](#) of the inter-transaction gap distribution
- **Target Variable**: Average monthly revenue in the target period (the prediction horizon, currently 3 months after cutoff).
- All the metrics per customer is a single measurement (R,F,M,C) in a lookback window. The setup is to predict the total revenue per customer in the target period.
- All features except consistency are log-transformed



One year of data

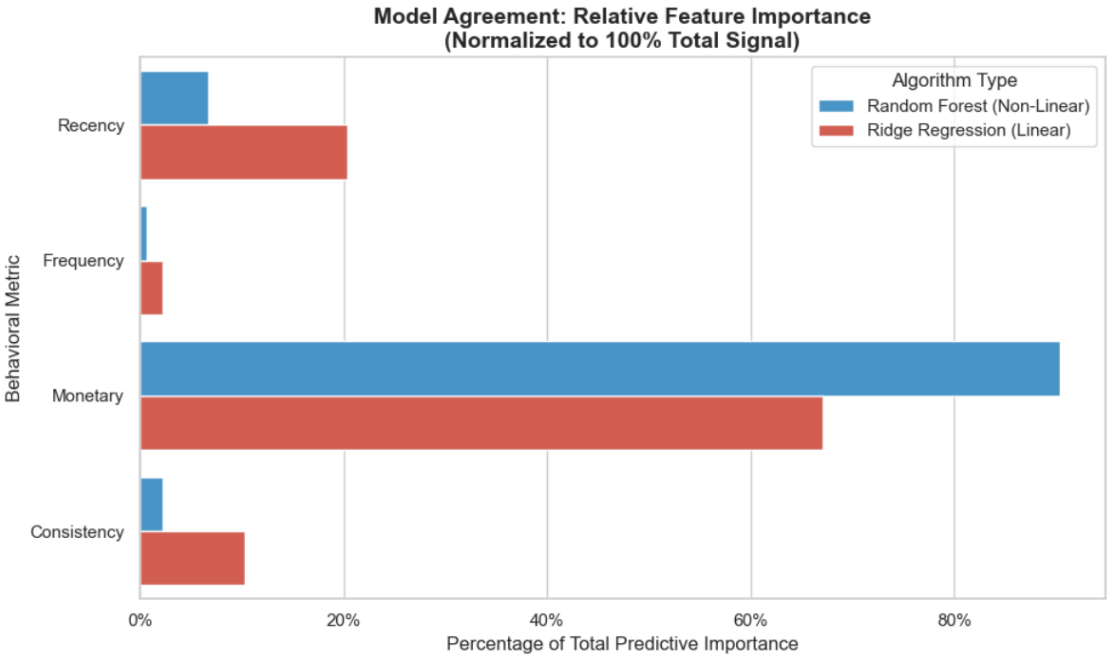
AccountID	Target_Revenue	Recency	Frequency	Monetary	Consistency
64031890-2D28-43E0-9504-5C57957CA4C8	0.019803	4.189655	1.791759	1.778336	0.733155
343B42F2-1CE2-43F4-B1EE-AC815DB84928	0.019803	4.955827	1.098612	1.924249	1.000000
A0726D41-9590-478B-BFBF-355201CDBC3E}	0.029559	4.727388	1.098612	2.099244	1.000000
C7CDA405-B1BE-4351-9BCF-186EBBF10A0E	0.067659	4.499810	2.397895	1.934416	0.457467
{33E2A8CE-FF80-4BF4-8156-517395F26BE6}	0.067659	4.077537	1.098612	1.695616	1.000000

Correlations



- Frequency, monetary and consistency are highly correlated

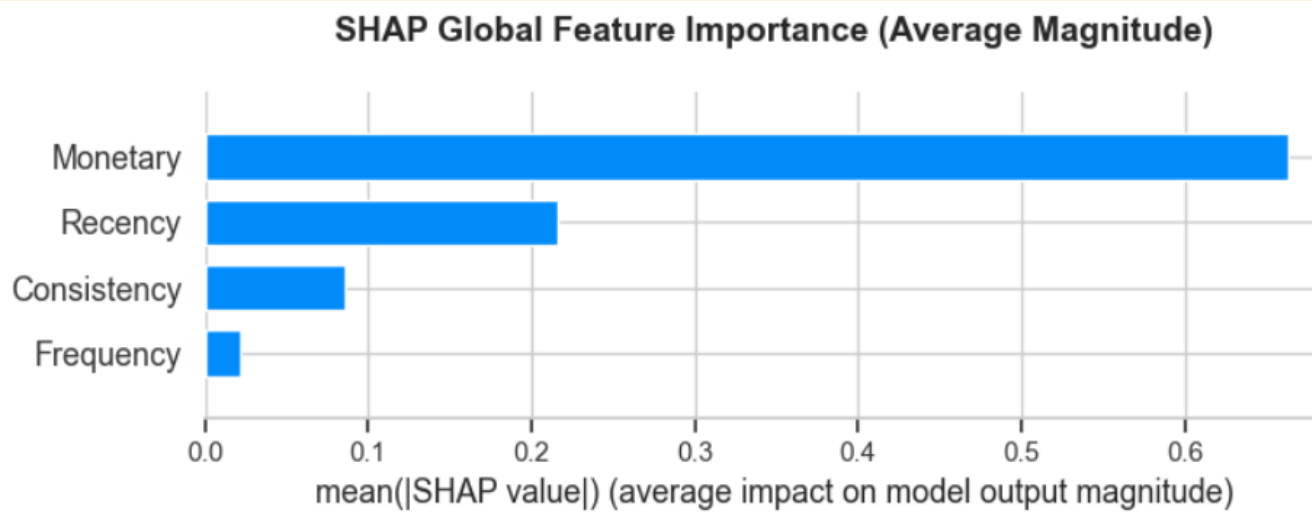
Feature Importance via Predictive Models



Tried three methods for feature selection:

1. Random forest
2. Ridge regression
3. XGBoost + Shapley values

All full models R^2 is about 0.6



Incremental model comparison – Best feature selection

Below all R^2 's are from out-of-sample data

- Average target revenue ~ Monetary : $R^2=0.54$
- ~ Monetary + Recency : $R^2=0.57$
- ~ **Frequency**+ Recency + : $R^2=0.51$
- ~ Monetary + Recency + Entropy : $R^2=0.57 + 0.001$
- ~ Monetary + Recency + log utility : $R^2=0.57 + 0.005$
- ~ Monetary + Recency + 2nd moment : $R^2=0.57 + 0.006$
- ~ Monetary + Recency + active weeks : $R^2=0.57+ 0.001$

Top-line takeaways

- Consistency features add very little to predicting the future revenue
- Monetary dominates
- Recency should stay

Behavior Features

What are the measures behind the three dimensions of behavior?

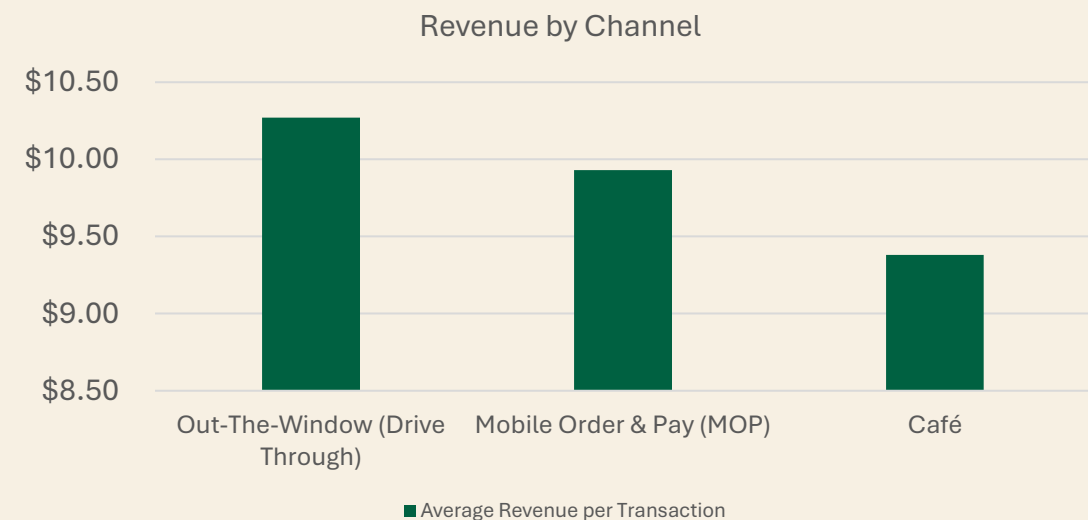
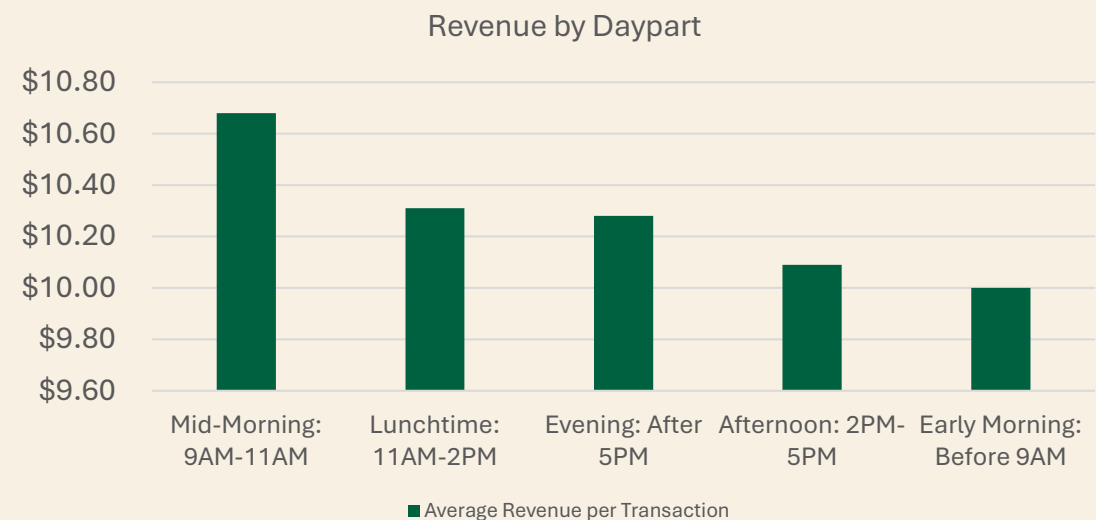
BEHAVIOR SIGNALS		
SR and Discount	Product Depth	Channel
Highest	Highest	Highest
High	High	High
Medium	Medium	Medium
Low	Low	Low
Low	Low	Low
N/A	N/A	N/A

- Easy to understand
- Ordered in an intuitive way*
- Intuitive so that stakeholders believe it
- Correlate or predict future value
 - Retention
 - Value
- Linked to strategic

Methodology: All at once feature selection approach.

- After the model with the RFMC features.
- Find the features that predict **90-day spend (revenue)**.
- Use exhaustive feature list
- Use feature selection approaches
- Use R-squared to explain how much of the 90-day spend is explained by the features

Behavior Features



We explored transaction data of a subset of customers and found directional correlations between average revenue per ticket and both daypart and channel usage. These correlations were statistically real (p-value near 0) and represent actionable levers for the business to utilize. Utilizing calculated scores for daypart and channel usage, we can assign each customer to a high/middle/low value state at any time grain and cadence*.

This part of the analysis was limited in the following ways:

- Starbucks Rewards members only
- Date range of March 10th – May 23rd 2026 to align with the start of Next Gen Loyalty
- Customers who had more than 2 transaction-days per week

*Normalized for transaction frequency to not overweigh customers with multiple transactions per day

Behavior Features

Starbucks Rewards Status Tiers

We found differences in average revenue per transaction between the 3 different loyalty tiers (Green/Gold/Reserve), but these differences are not actionable nor informative for the business independent of transaction frequency.

Similarly, we found increased weekly value generated by customers who redeemed Star Rewards when making transactions, but this is necessarily true because customers would have generated more revenue to obtain the Stars before transacting.

	Average Ticket Revenue	Average Frequency
Green	\$9.80	1.49 transactions/week
Gold	\$9.65	2.17 transactions/week
Reserve	\$9.67	4.16 transactions/week

Takeaway

Starbucks Loyalty and Rewards data are not independently useful for segmenting customer value states because they are inherently tied to transaction frequency. The data may become informative when we combine the behavioral and RFM features, but currently it does not contribute information to the model that is not already included in other inputs (directly or indirectly).

Feature List: (in [github](#))

State Methodology

How do we build the States?

I have the behaviors I want to include and the features that describe those behaviors, how do I identify the High, medium, low and, eventually, the State definitions?

State Requirements

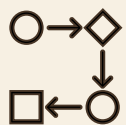
What makes effective States



- The segments should separate customers by their **current value** and be ordered. The customers within a segment should be similar in value while customers in different segments should differ.

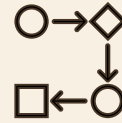


- The segments should be **reasonably sized**; both in terms of the number of customers within the segments and the amount of spend the segment generates. The sizes of the segments should be *stable through time*.

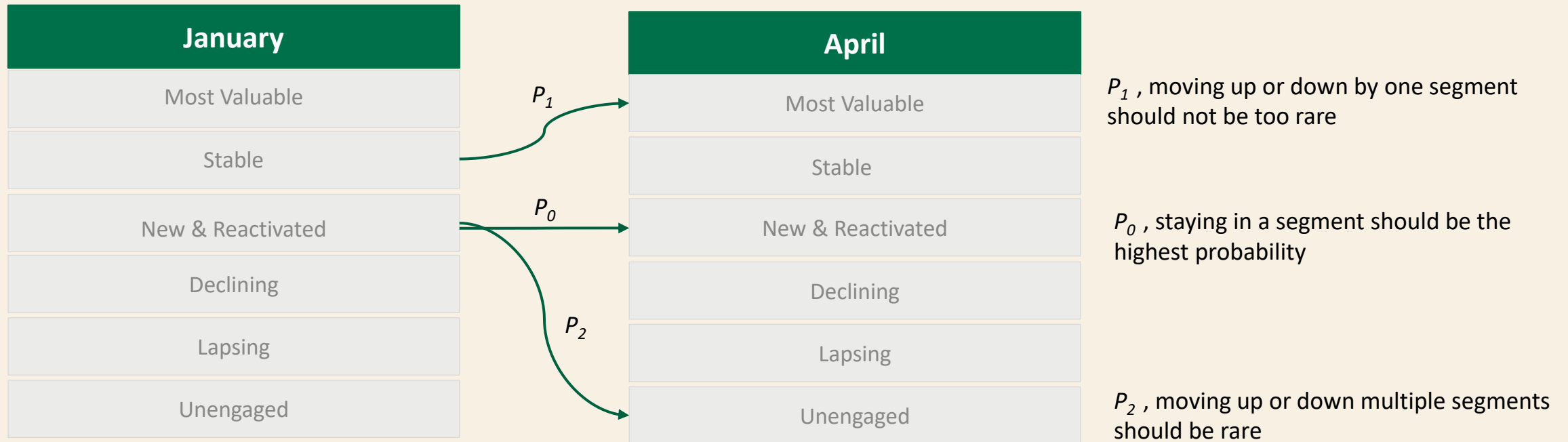


- **Transition probabilities** should be stable through time and not too small or big. Transition probability to stay in a segment should be larger than the transitions.

Transition probabilities



Customer Migrations should be smooth, not too jumpy.



These requirements will be highly dependent on the time-windows we use to define the states.

Success Metrics

We will evaluate each States we create with these metrics

Validation metrics are in a Notebook:
[state_validation.ipynb](#)

Because there are several validation metrics, and we don't know what is the optimum value achievable, we'll start with a baseline, simple approach and improve from there.

Approach for Defining Behaviors

If we take the *conceptual behavior approach*, we still need to define the metrics that are used to place customers into States.

- To develop the behavior measures, we will use variables that describe that behavior in a *logical* way and are correlated with future value. These are the *features*.
- The approach will be supervised by running *feature selection* algorithms, that *account for* RFMC behaviors prior to selecting the behavioral features.
- Once we decide on the features that should measure the behavior, we will use a simple regression to determine the relationship between those features and future value.
- The output will be a linear equation that links the conceptual behavior to future value via these features.

Example is here: [CustomerStatesEDA/behaviorApproach.md](#) at main · jcourboi/CustomerStatesEDA

Final States Definitions

Defining the H/M/L categories

Once we have the measures for RFMC and each behavior, we have to

- identify the cut-points that define the H/M/L categories
- Collapse all possible combinations to ~20 states

Find the best solution that optimizes the success metrics.

We don't have an algorithm for this yet, and there might not be one, but we will be working on this next.

State	State	VALUE SIGNALS		BEHAVIOR SIGNALS		
		Monetary	Recency	SR and Discount	Product Depth	Channel
1	Most Valuable	Highest	Highest	Highest	Highest	Highest
2						
3						
4	Stable	High	High	High	High	High
5						
6						
7	New & Reactivated	Low	High	Medium	Medium	Medium
8						
9						
10	Declining	Medium	Medium	Low	Low	Low
11						
12						
13	Lapsing	Low	Low	Low	Low	Low
14						
15						
16	Unengaged	N/A	N/A	N/A	N/A	N/A
17						
18						
19						
20						

Predictive Models

Predictive Models

The predictive model work is slated for July. However, notice that the models are highly tied to the State definitions.

MODEL	APPROACH	USE CASES
Transition Probabilities <i>What is the probability of moving to another State?</i>	• Markov models • Force rare transitions to zero • Evaluated using both prediction accuracy and longitudinal stability	• Predictions used to calculate value of State migrations (ROI). • Prioritize State transition efforts • Identify customers with the greatest growth potential • Identify customers at retention risk
Lifetime Value <i>What is the future lifetime value of a customer over a defined planning horizon?</i>	• State transitions X lifetime value of states • Evaluated using forward looking accuracy • State level aggregate forecasts should remain stable through time.	• Calculate value of State migrations (ROI). • Prioritize State transition efforts • Identify customers with the greatest growth potential • Identify customers at retention risk

Notice that this is two different levels of modeling, State and Customer, of which the methodology may be different (customer level modeling requires including customer-level features).

Predictive Model Scope

Each model will have a customer-scope and a State-scope:

SCOPE	APPROACH	USE CASES
State <i>What is the “average” value of a customer in this state and the transition probabilities?</i>	Average of customer predictions in States?	- Predictions used to calculate value of State migrations (ROI). - Prioritize State transition efforts
Customer <i>What is each customers’ LTV and transition probability</i>	Transitions and LTV based on customer-level features.	- Identify customers with the greatest growth potential - Identify customers at retention risk

What is Next

Next Steps

- Detailed timeline needs to be revised given that we have identified an approach.
- I would like to meet on a regular basis to have this type of meeting. We need a week to get some stuff done, but after that I would like to start **reviewing results weekly**
- Meet with **stakeholder teams** to introduce them to the approach and get their feedback – already set up
- Detailed description of the predictive models

Feature Tracker

Customer Feature Development Status

Feature	Description	Status
RFMC (value)	Descriptions of customer value	EDA in progress
SR engagement	Engagement with SR features (stars)	EDA in progress
Day part preferences		EDA in progress
Product preferences	Category depth, LTO engagement	EDA in progress
Channel use	Multiple channel use	EDA in progress
Pricing/value choices	Average order value	Not started
Promotion / game use	Overlap with SR engagement	Not started

Loss functions

States are homogeneous
and different from each
other

Within and between sum of
squares (ANOVA) –
90-day revenue?

States are sorted by value

??

States are stable through
time – customers migrate
but not too fast